Budgeted Heterogeneous Ensembles for Robust OCR*

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Abstract. Optical character recognition remains brittle under document heterogeneity: systems that work well on dense printed forms often fail on handwriting, tables, or noisy scans. We study whether this fragility can be reduced by scaling *test-time computation* rather than retraining a larger OCR model. We present **FOREST** (Framework for Reasoning and Ensemble Selection at Test-time), a modular inference-time framework that treats heterogeneous OCR engines as parallel hypothesis generators. Given a document image, FOREST allocates a fixed call budget across workers and augmentations, aligns region-level outputs with IoU-based Hungarian matching, scores competing candidates with a cluster-wise Transformer verifier, and uses disagreement features for both final selection and abstention.

On OCRBench v2, FOREST improves the strongest single worker from 14.8% CER to 10.6% at the default operating point (C=8), and further to **8.4%** at the largest tested budget (C=16), corresponding to a 43.2% relative reduction. At matched budget, it outperforms majority voting, confidence-weighted voting, Deep Ensembles, and stacking built on the same worker pool. The resulting uncertainty estimates are actionable: under selective prediction, FOREST reaches **5.8%** CER on 85.7% of accepted samples. Cross-dataset results on DocBank and ICDAR2013, together with ablations and calibration analyses, show that spatial alignment, learned verification, and disagreement modeling all contribute to the gains. These results position FOREST as a practical route to test-time scaling for robust OCR without modifying the underlying OCR engines.

Keywords: OCR · ensemble learning · test-time computation · uncertainty quantification · selective prediction

1 Introduction

Optical character recognition (OCR) underpins document digitization, accessibility tooling, and downstream information extraction, yet deployment-time robustness remains unresolved. Systems that work well on printed forms often degrade on handwriting, tables, or low-quality scans; conversely, recognizers that handle handwriting well can struggle on layout-heavy pages. In practice, this forces brittle routing rules, domain-specific fine-tuning, or a single over-provisioned OCR stack.

* Code, training logs, and supplementary material will be released upon acceptance.

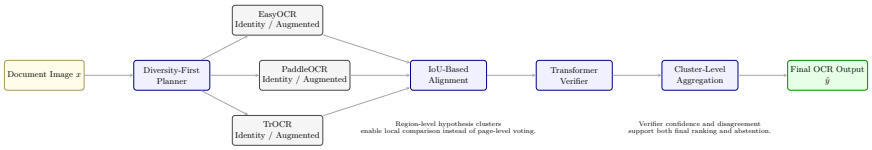


Fig. 1: Overview of FOREST. A fixed call budget is allocated across heterogeneous OCR workers, aligned into region-level clusters, scored by a cluster-wise verifier, and merged into the final OCR output.

This paper studies a different lever: *test-time computation*. Instead of committing to one OCR engine, we spend a fixed inference budget on multiple heterogeneous workers and then reason over their outputs to produce a better final prediction. This perspective is related to recent work on inference-time scaling and adaptive computation [25,10,3], but OCR must combine structured outputs: partially overlapping text strings, bounding boxes, and confidences that are not naturally aligned.

We propose **FOREST** (**F**ramework for **R**easoning and **E**nsemble **S**election at **T**est-time), a budgeted inference-time framework for robust OCR. FOREST allocates calls across workers and augmentations, aligns outputs into region-level hypothesis clusters via IoU-based Hungarian matching, scores competing candidates with a learned Transformer verifier, and combines verifier confidence with cross-worker disagreement for both final prediction and selective abstention. The central design choice is local reasoning: disagreements are resolved at the cluster level, where complementary workers often expose different failure modes on the same region.

Contributions. We make three contributions. First, we formulate robust OCR as a budgeted reasoning problem over heterogeneous workers rather than a single-model robustness problem. Second, we introduce an alignment-verification-aggregation pipeline that turns inconsistent OCR outputs into comparable region-level candidates. Third, experiments on OCRBench v2 and additional transfer benchmarks show clear matched-budget gains and useful abstention behavior: at C=8, FOREST reduces CER from 14.8% to 10.6%, at C=16 it reaches **8.4%**, and selective prediction reduces accepted-subset CER to **5.8%**.

2 Related Work

Test-time computation and learned combination. Additional inference-time compute can improve robustness through adaptive computation, augmentation, ensembling, or learned combination [25,10,3,26,5,27,18,7,8]. FOREST is closest in spirit to stacking and ensemble-based uncertainty estimation, but it operates on structured OCR outputs rather than aligned class scores. This makes spatial alignment and region-level reasoning necessary before any meta-learner can be effective.

OCR and uncertainty. Modern OCR systems span recognizers such as CRNN, TrOCR, PaddleOCR, EasyOCR, and MMOCR [24,2,19,6,16]. Document models increasingly incorporate layout and multimodal cues through LayoutLM, LayoutLMv3, and broader document benchmarks and models [28,12,13,15]. FOREST is complementary to this line of work: it keeps existing OCR engines fixed and learns how to reconcile their text, geometry, and confidence signals at inference time. Our selective-prediction component is motivated by prior work on calibration, uncertainty estimation, abstention, and conformal reliability control [11,23,18,22,9,1], but applies these ideas to aligned OCR hypotheses rather than a single categorical prediction.

3 Problem Setup

Structured OCR output. Let $x \in \mathcal{X}$ denote a document image and let the target output be a set of text-box pairs $y^\star = \{(t_i^\star, b_i^\star)\}_{i=1}^{N^\star}$, where each $t_i^\star$ is a text string and $b_i^\star \in \mathbb{R}^4$ is its bounding box. A worker $w_k \in \mathcal{W}$ produces a structured prediction $\hat{y}_k = \{(t_{k,j}, b_{k,j}, p_{k,j})\}_{j=1}^{N_k}$, with worker-specific confidences $p_{k,j} \in [0,1]$.

Budgeted inference. FOREST operates under an explicit *call budget* C, the total number of worker invocations permitted for one input image. Instead of committing to a single OCR engine, the system constructs an inference schedule

$$\mathcal{S}(x, C) = \{(w_i, a_i, t_i)\}_{i=1}^{C}, \tag{1}$$

where each entry specifies a worker identity, an optional image augmentation, and, when needed, a tile or crop. This schedule is the main control knob that trades latency for accuracy. Small budgets correspond to practical low-latency use, while larger budgets expose the test-time scaling behavior of the framework.

Hypothesis pool and objective. Given x and budget C, the system produces a hypothesis pool

$$\mathcal{H}(x) = \{h_i = (y_i, p_i, m_i)\}_{i=1}^{M}, \tag{2}$$

where y_i is a structured OCR hypothesis and m_i stores metadata such as worker identity, augmentation, and geometric context. The goal is to compute a final structured output $\hat{y}$ that minimizes expected OCR error while preserving calibrated uncertainty:

$$\hat{y} = \arg\min_{y \in \mathcal{Y}} \mathbb{E}[\mathrm{CER}(y, y^\star) \mid \mathcal{H}(x), C]\,. \tag{3}$$

Calibration requirement. Accuracy alone is insufficient in high-stakes OCR pipelines. We also require that higher disagreement correspond to higher empirical error, so that uncertain samples can be deferred for human inspection. This motivates the epistemic-uncertainty component of FOREST and the selective-prediction evaluation in Section 5.

4 FOREST Framework

FOREST converts a fixed inference budget into a pool of region-level OCR hypotheses and resolves them at the cluster level rather than the page level. The pipeline has four stages: budgeted schedule construction, spatial alignment, candidate verification and aggregation, and disagreement-aware uncertainty estimation.

4.1 Budgeted Hypothesis Generation

Given an input document x and a call budget C, the planner selects C worker-augmentation pairs $\mathcal{S} = \{(w_i, a_i)\}_{i=1}^{C}$. In our implementation, the planner follows a diversity-first heuristic: it allocates at least one call to each OCR engine before reusing the same engine with additional augmentations. This scheduling rule is simple but important. Diverse workers expose complementary failure modes early, which produces more informative disagreement patterns than repeatedly querying a homogeneous recognizer.

The union of all worker outputs defines a page-level hypothesis pool

$$\mathcal{H}(x) = \bigcup_{i=1}^{C} f_{w_i}(a_i(x)), \tag{4}$$

where each element contains a text string, bounding box, worker confidence, and source metadata. The rest of FOREST reasons over this hypothesis pool.

4.2 Structured Alignment with Hungarian Matching

Raw OCR outputs cannot be compared directly because different workers may place slightly different boxes around the same text region. FOREST therefore aligns hypotheses into semantic clusters before any voting or learned scoring. For candidate boxes b_i and b_j, we define the matching cost

$$\mathrm{Cost}(i, j) = 1 - \mathrm{IoU}(b_i, b_j), \tag{5}$$

and solve the resulting assignment problem with Hungarian matching [17]. We retain links whose IoU exceeds $\tau_{\mathrm{iou}} = 0.5$ and create a new cluster for unmatched candidates. The resulting clusters $\mathcal{C} = \{C_1, \ldots, C_M\}$ represent semantic text regions with several competing OCR interpretations. This step converts OCR from a page-level ensembling problem into a local selection problem.

4.3 Consistency-Aware Verifier and Aggregation

For each hypothesis $h_{m,i}$ inside cluster C_m, we construct a 32-dimensional feature vector that captures four signal groups: geometry, text agreement, worker identity,

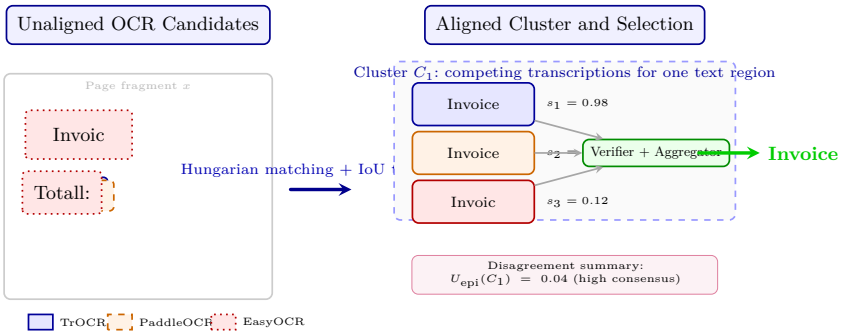


Fig. 2: FOREST alignment and verification. Workers propose OCR candidates, Hungarian matching forms semantic clusters, and the verifier scores candidates within each cluster while disagreement defines epistemic uncertainty.

and intrinsic confidence. These vectors are stacked into a cluster matrix $\mathbf{F}_m$ and processed by a two-layer Transformer:

$$\mathbf{H}_m = \text{Transformer}(\mathbf{F}_m). \tag{6}$$

The verifier outputs one scalar score per candidate,

$$s_{m,i} = \sigma(\mathbf{w}^\top \mathbf{h}_{m,i}), \tag{7}$$

which estimates whether the candidate is the most reliable member of the cluster. We train the verifier and the final aggregator with region-level supervision: after a cluster is matched to a reference text region, the candidate with the lowest normalized edit distance to the reference transcription is marked positive and the remaining cluster members are marked negative. A Random Forest aggregator [4] then ranks cluster members using verifier scores together with the structured features and selects the final cluster prediction. We choose this lightweight nonlinear ranker because the final feature set is low-dimensional, heterogeneous, and trained with limited supervision; in this regime, a heavy second-stage neural model is unnecessary.

4.4 Epistemic Uncertainty

FOREST uses disagreement not only for abstention but also as part of final ranking. For a cluster C_m, we define disagreement as the average normalized pairwise edit distance:

$$U_{\text{epi}}(C_m) = \mathbb{E}_{i \neq j}\left[\frac{\text{EditDist}(y_i, y_j)}{\max(|y_i|, |y_j|)}\right]. \tag{8}$$

Page-level uncertainty is the mean cluster uncertainty,

$$\bar{U}_{\text{epi}} = \frac{1}{M}\sum_{m=1}^{M} U_{\text{epi}}(C_m), \tag{9}$$

and the final abstention score combines disagreement with verifier confidence:

$$U_{\text{total}}(x) = \alpha \bar{U}_{\text{epi}} + (1-\alpha)\left(1 - \frac{1}{M}\sum_{m=1}^{M} \max_i s_{m,i}\right). \tag{10}$$

The same disagreement statistics are appended to the aggregator features, so disagreement modeling affects both final prediction and selective abstention. At deployment time, the system abstains when $U_{\text{total}}(x) > \theta_U$.

5 Experiments

5.1 Experimental Setup

Datasets and protocol. We report main results on *OCRBench v2* [21], which contains 1,809 heterogeneous evaluation images spanning printed forms, tables, handwriting, and noisy scans. To test whether the gains persist beyond the main benchmark, we also evaluate the default configuration on *DocBank* [20] and *ICDAR2013* [14]. We use CER as the primary metric and additionally report WER, latency, calibration, and coverage under selective prediction.

Workers, training, and baselines. The worker pool contains PaddleOCR [6], EasyOCR, and TrOCR [19]; the default augmentation set includes identity, sharpening, denoising, contrast enhancement, and Otsu binarization. Under the default C=8 operating point, the planner first allocates one call to each engine and spends the remaining calls on augmented views to preserve cross-engine diversity. The verifier is trained on 10,000 OCRBench training samples using AdamW with learning rate 3×10^{-4}, batch size 32, and 10 epochs, with labels derived from matched ground-truth text regions as described in Section 4. We compare against the single best worker, majority voting, confidence-weighted voting, Deep Ensembles [18], and stacking [27]. All ensemble baselines use the same worker pool and call budget as FOREST; because OCR outputs are structured, they operate on the same aligned clusters and differ only in the cluster-level scoring rule.

5.2 Main Results and Diversity Effect

Table 1 shows that FOREST improves the strongest single worker from 14.8% CER to 10.6% at the default operating point (C=8), and to **8.4%** at C=16. The matched-budget comparison at C=8 confirms that the gain does not come from extra compute alone: FOREST outperforms majority vote, confidence-weighted voting, Deep Ensembles, and stacking at the same budget.

Why diversity matters. Table 2 isolates the source of the gain at fixed budget. Using three heterogeneous engines is consistently better than spending all eight calls on one engine with more augmentations, which shows that the benefit comes from complementary failure modes rather than denser search around a single recognizer.

Table 1: Main OCR results on OCRBench v2. CER and WER are lower-is-better. Rows above the divider compare matched-budget baselines at C=8 using the same worker pool and aligned clusters; rows below show the FOREST budget sweep.

Method	**Budget** C	**CER↓**	**WER↓**	**Latency (s)**
Single best (TrOCR)	1	14.8%	20.2%	0.50
Majority vote	8	12.4%	17.3%	2.80
Conf.-weighted vote	8	11.2%	16.8%	2.80
Deep Ensembles [18]	8	11.1%	16.4%	2.80
Stacking [27]	8	11.3%	16.5%	2.80
FOREST (budget sweep):				
$C = 2$	2	13.5%	19.1%	0.90
$C = 4$	4	11.9%	17.4%	1.60
$C = 8$	8	10.6%	15.8%	2.80
$C = 16$	16	**8.4%**	**12.5%**	5.10

Table 2: Agent diversity vs. search density on OCRBench v2 at budget C=8. "Dense" uses a single engine across eight calls, while "Diverse" spreads the same budget across heterogeneous workers.

Configuration	**Composition**	**CER↓**	**WER↓**
Dense (Paddle)	1 engine, 8 calls	13.5%	18.8%
Dense (EasyOCR)	1 engine, 8 calls	14.1%	20.4%
Dense (TrOCR)	1 engine, 8 calls	13.2%	18.2%
Diverse (Heterogeneous)	**3 engines, shared 8-call budget**	**10.6%**	**15.8%**

5.3 Compute–Accuracy Scaling

Figure 3 plots CER against call budget. The gains are monotonic across the tested budgets, with the largest marginal improvement appearing between C=1 and C=8. The curve then exhibits diminishing returns, which is consistent with the idea that extra calls mainly refine a smaller set of ambiguous regions once the worker pool has already exposed most disagreements.

5.4 Generalization and Robustness

The default C=8 configuration transfers consistently across external datasets (Table 3): FOREST reduces CER from 14.8% to 10.6% on OCRBench v2, from 12.5% to 8.7% on DocBank, and from 17.2% to 11.4% on ICDAR2013. The gains therefore are not confined to one benchmark or one document style.

5.5 Ablation Results

Ablations at C=8 (Table 4) quantify the contribution of each module. Removing **spatial alignment** causes the largest regression (+3.6 CER points), because

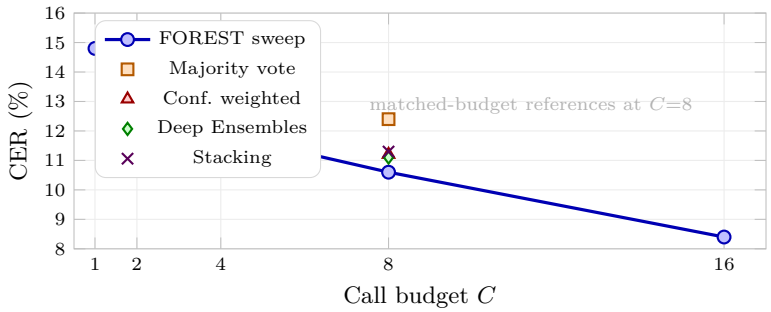


Fig. 3: Compute–accuracy tradeoff on OCRBench v2. The blue curve shows the FOREST budget sweep; colored markers at C=8 denote matched-budget baselines built on the same worker pool.

Table 3: Cross-dataset validation of the default FOREST configuration (C=8). FOREST maintains consistent gains across document-centric and scene-text benchmarks.

Dataset	**Single Best**	**FOREST** $C = 8$	**Rel. Improvement**	**Sample Count**
OCRBench v2	14.8%	10.6%	28.4%	1,809
DocBank	12.5%	8.7%	30.4%	500
ICDAR2013	17.2%	11.4%	33.7%	1,095
Average	14.8%	10.2%	31.1%	-

the system can no longer reliably group hypotheses into semantic regions before selection. Removing the **learned verifier** adds +0.9 CER points, while removing **disagreement features** from both the final ranker and the abstention score adds +2.2 points. Replacing diversity-first planning with uniform worker reuse causes a smaller but consistent regression (+0.2), indicating that schedule quality matters even under a fixed budget.

Table 5 shows where the gain is concentrated. The largest improvements appear on handwriting and tables, where complementary worker biases and local verification are most useful.

5.6 Calibration and Selective Prediction

Figure 4 evaluates whether disagreement tracks OCR risk. The left panel shows a strong monotonic relationship between mean epistemic uncertainty and CER (Pearson r=0.91). Table 6 reports the same effect numerically: FOREST achieves an ECE of 0.021 and reduces accepted-subset CER from 10.6% to **5.8%** at θ_U=0.6 while retaining 85.7% coverage. This is useful in practice because difficult pages can be deferred instead of being forced through the OCR pipeline with overconfident predictions.

Table 4: Ablation study on OCRBench v2 at $C = 8$. Each row removes one component from the full FOREST pipeline. ΔCER is the absolute increase in CER.

Variant	CER↓	ΔCER
FOREST (full)	**10.6%**	—
w/o Spatial alignment	14.2%	+3.6
w/o Learned verifier	11.5%	+0.9
w/o Disagreement features	12.8%	+2.2
w/o Diversity-first planning	10.8%	+0.2
w/o All (majority voting)	12.4%	+1.8

Table 5: Per-document-type CER (%) ablation. Gains are largest on handwriting and tables.

Variant	Print.	Handwr.	Tables	Forms	Overall
FOREST (full)	6.1%	15.2%	10.4%	9.8%	10.6%
w/o Spatial align.	9.8%	20.4%	16.5%	13.2%	14.2%
w/o Verifier	7.2%	16.8%	11.9%	10.7%	11.5%
w/o Disagreement	8.3%	18.4%	13.1%	11.8%	12.8%
w/o Diversity planning	6.5%	15.7%	10.8%	10.1%	10.8%
Majority vote	8.9%	19.1%	13.8%	12.3%	12.4%

6 Discussion

The experiments support a simple interpretation of FOREST: extra test-time compute helps OCR when it is spent on genuinely diverse workers and paired with region-level verification. The gain is therefore not just a consequence of averaging more outputs. It comes from exposing complementary errors and resolving them locally after alignment, which is exactly where page-level voting is weakest. The compute curve also shows diminishing returns beyond the default C=8 operating point, suggesting a natural deployment split between low-latency settings and higher-budget offline processing.

Practical scope. FOREST is best viewed as a reusable coordination layer on top of existing OCR systems. It keeps the underlying recognizers unchanged, benefits from stronger worker pools as they become available, and adds a reliability signal that can support human deferral in production pipelines. The main practical tradeoff is additional inference cost relative to a single OCR engine, which makes the default C=8 operating point a useful compromise between accuracy and latency. In specialized domains, lightweight recalibration of the verifier is a natural next step rather than a redesign of the framework.

7 Conclusion

FOREST turns a fixed OCR inference budget into a structured reasoning process over heterogeneous workers. Across matched-budget comparisons, transfer

Table 6: Calibration on OCRBench v2. Lower ECE and Brier score indicate that FOREST's disagreement-aware uncertainty is better aligned with empirical error.

Method	ECE↓	Brier↓	Coverage	CER (accepted)↓
Single model (TrOCR)	0.121	0.210	100%	14.8%
Confidence-weighted vote	0.083	0.174	100%	11.0%
FOREST (ours)	**0.021**	**0.088**	85.7%	**5.8%**

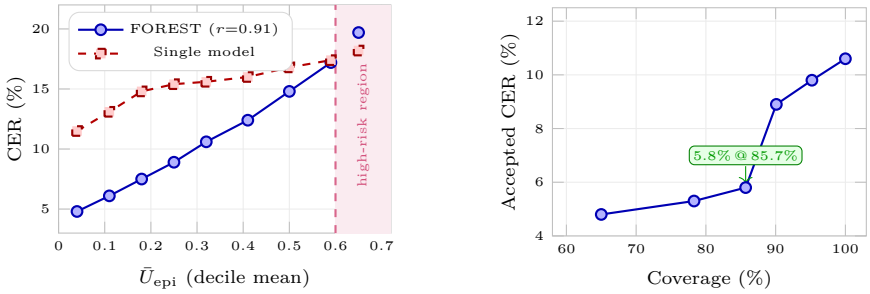


Fig. 4: Uncertainty calibration. Left: mean epistemic uncertainty tracks CER. Right: selective prediction lowers accepted-subset CER to **5.8%** at 85.7% coverage.

benchmarks, and calibration analyses, the main empirical finding is consistent: budgeted local reasoning beats page-level combination for robust OCR. On OCRBench v2, CER drops from 14.8% for the strongest single worker to 10.6% at C=8 and **8.4%** at C=16, while selective prediction reduces accepted-subset CER to **5.8%**. These results suggest that lightweight meta-reasoning is a practical path to more robust OCR without retraining larger base recognizers.

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